UNIVERSITY of WASHINGTON

Low-Power Earring for Longitudinal Temperature Sensing

Shirley Xue*[1], Yujia (Nancy) Liu*[2], Vikram Iyer[1], Shwetak Patel[1]

* Equal Contribution

[1] Paul G. Allen School of Computer Science & Engineering

[2] Electrical & Computer Engineering

Smart Earrings

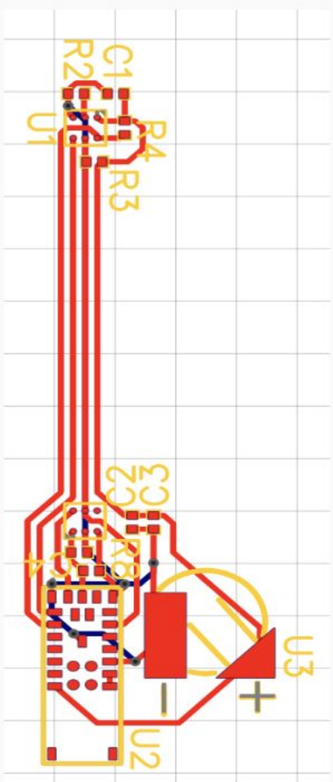
with Temperature sensing

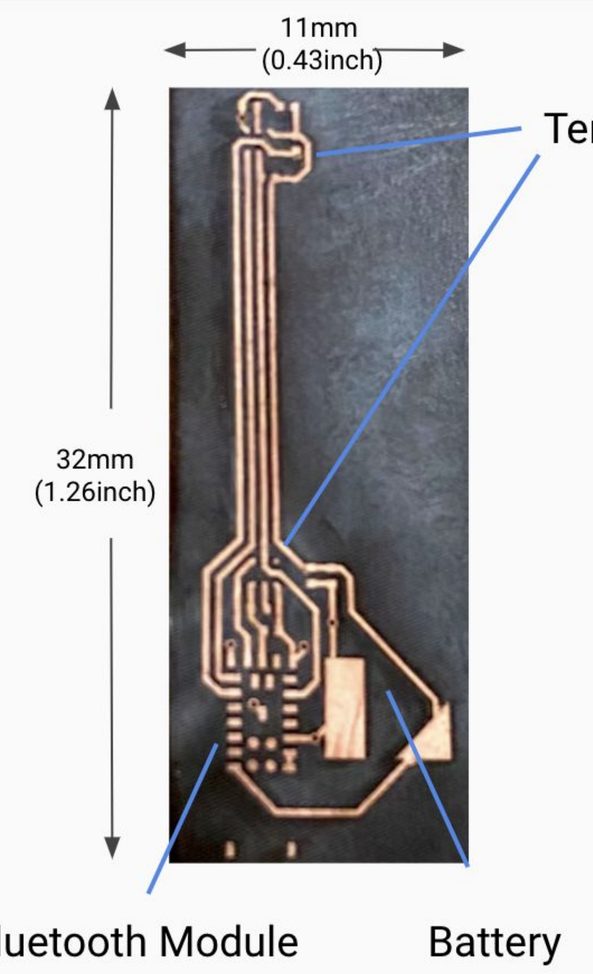
Long-term Passive Sensing

System Design

- Small scale
- Low-power
- Light-weight

- Power harvesting
  - Solar cell
  - Magnetic

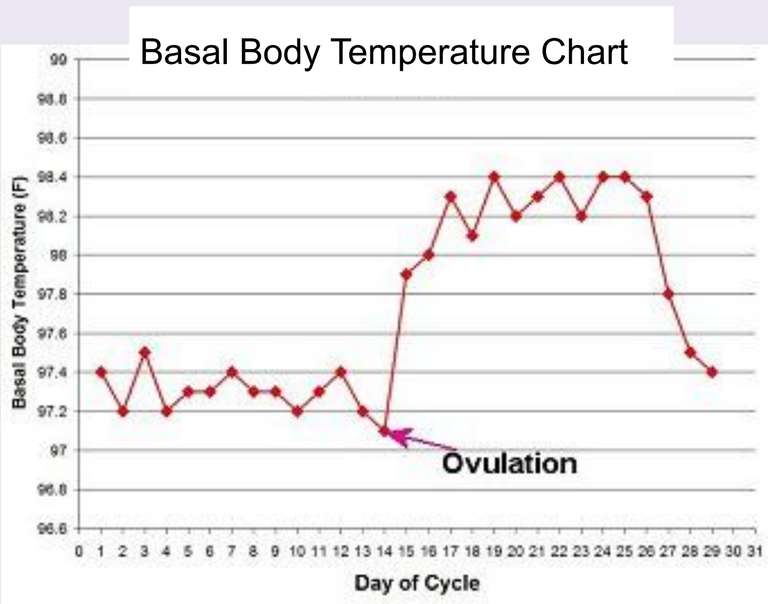




Temp Sensor	HCD2010
Bluetooth Module	nRF52832
Battery	ML414H-IV01E

Applications

Menstrual Cycle



Emotion Detection